

Differential effects of forestry plantations on bird diversity: A global assessment

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ABSTRACT

Planted forests are becoming increasingly common worldwide, replacing natural forests and compromising their associated biodiversity. However, not all forest plantations are the same. Aiming to identify key plantation features that could be managed to improve bird diversity, we conducted a meta-analysis examining the differential effects of several kinds of plantations on bird species richness and abundance. Our literature survey provided 123 case studies (68 cases for species richness and 55 for abundance), which showed that, although forest plantations have negative effects on both richness and abundance, there is a large heterogeneity of responses among plantation types. The least negative effects were found at plantations established with native tree species and mixed arrangements, along with those intended for protective uses. Moreover, exotic monocultures for commercial purposes were found to be the most negative plantations for bird diversity. Likewise, small (< 36 ha) stands, those with large rotation times, as well as plantations connected with native forest remnants at the landscape level were more avifauna-friendly. However, the presence of understory vegetation and the lack of management were not beneficial to bird diversity, as was also previously suggested by local studies. These global patterns of change in bird diversity in forest plantations hint which features could be managed to reduce their negative effects on bird diversity, making these productive lands friendlier to biodiversity in the long-term.

1. Introduction

Planted forests constitute about 7% (278 million ha) of the total forested area worldwide, with an annual increase rate of 3.3 million ha during the 2010–2015 period (FAO, 2015; Keenan et al., 2015). Of this forested area, 55% is located in temperate areas and 22% in tropical areas, which have experienced an increase of 67% and 51%, respectively, over the last 25 years (FAO, 2016). Forest plantations are an important component of the global timber industry, contributing in different ways to the economy of many countries in Asia, South America, and Europe (FAO, 2001; ITTO, 2016). In this sense, forest plantations provide multiple goods to the society (e.g., raw materials for commercial wood processing), strengthen local economies, and generate employment. Also, these plantations can provide several ecosystems services (e.g., carbon sequestration) and can be used to recover degraded lands in ecological restoration programs (Baral et al.,

2016; Bull et al., 2006; Carle et al., 2002; FAO, 2016, 2018; Parrotta and Knowles, 1999; Silva et al., 2018) at both global (e.g., Bonn challenge intending to recover 150 million ha by 2020 and 350 million ha by 2030) and regional (e.g., Initiative 20 × 20 aiming to recover 20 million ha by 2020 in Latin American and Caribbean countries) scales.

For all of these reasons, local governments have promoted the expansion of forest plantations through national policies (fiscal, economic and financial incentives, along with legal guarantees for forestry companies and the emerging green bonus market) or international agreements, such as the Kyoto protocol (Bull et al., 2006; FAO, 2018; Masiero et al., 2015; UNFCCC, 2013). Therefore, it is expected that the area occupied by planted forests will increase in the forthcoming years (FAO, 2016). Such an increase in planted areas will result in land use changes, which, in some cases, represent the conversion of native forest habitats to productive plantations (INDUFOR, 2012). However, our knowledge on the effects of forest plantations on biodiversity is

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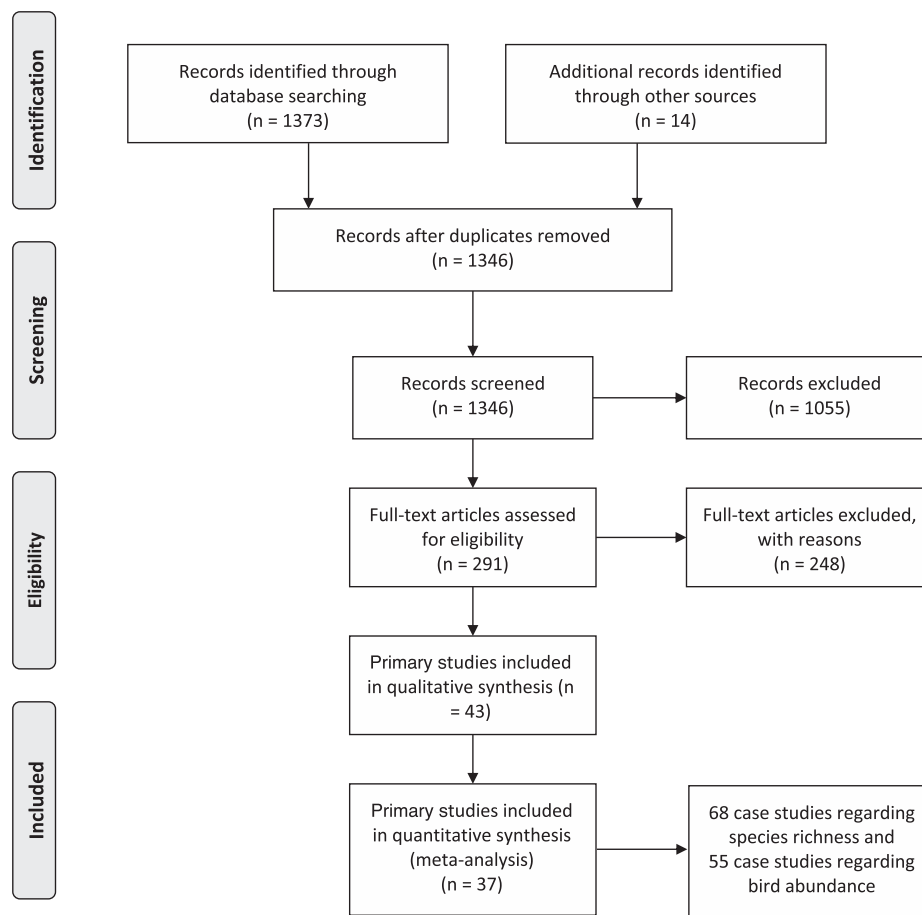


Fig. 1. PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analysis) flowchart of this study, summarizing the information selection sequence.

fragmentary and global change patterns have been little explored so far.

Planted forests are known to have detrimental effects on biodiversity and ecosystem services (Jones et al., 2018; Moreno-Mateos et al., 2017). Therefore, many studies conducted across different ecoregions have shown that forest plantations support an impoverished subset of the native avifauna (i.e., disturbance tolerant and generalist species) found in natural forests (Barlow et al., 2007; Calviño-Cancela, 2013; Felton et al., 2016; Martin-Garcia et al., 2013). Meanwhile, other studies found similar species richness with important species turnover in both planted and natural forests (Castaño-Villa et al., 2014a; Cid and Caviedes-Vidal, 2014; Farwig et al., 2009; Hsu et al., 2010). Regardless of the contrasting results found among local studies, it has been suggested that some local- and landscape-level features are able to reduce the negative effects of forest plantations on bird diversity. Thus, some studies have established that several factors, such as age, rotation time, tree density, degree of landscape connectivity of the plantation, understory type, number of species planted (mixed or monospecific plantations), and the origin of the planted species (native or exotic) could be largely determining local bird diversity (Bett et al., 2016; Castaño-Villa et al., 2014a; Cid and Caviedes-Vidal, 2014; Farwig et al., 2009; Rey-Benayas et al., 2010). Despite the value of plantation forest on biodiversity conservation has been acknowledged in the past decade (Brockerhoff et al., 2008), we lack a global and quantitative assessment to determine which plantation features could be managed to reduce the negative impacts of forest plantations on bird diversity. Therefore, it is of paramount importance to acknowledge the key management factors that could be a fundamental tool for policy-makers and wildlife managers. This is particularly significant for developing countries with high bird diversity, where extensive plantations are often set without proper management plans (Carle et al., 2002). Further, these factors can be

highly informative to improve forest restoration initiatives (Crouzeilles et al., 2016; Meli et al., 2017).

We conducted a global meta-analysis to synthesize the available literature on bird diversity in forestry plantations, with two objectives: (1) to assess the effects of forestry plantations on bird richness and abundance compared to native forests; and (2) to identify the key factors that could be managed in forestry plantations to improve bird species richness and abundance. Given that bird diversity responds to habitat heterogeneity and complexity, we expect forestry plantations to have less bird richness and abundance than native forests. Also, we expect some plantation features (e.g., origin of the planted species, management type, rotation time) to increase bird species richness and abundance, as these plantations may better resemble a natural forest habitat. We also provide management guidelines for making forest plantations avifauna-friendly.

2. Methods

2.1. Literature survey and data inclusion criteria

We searched the available literature by topic, using the ISI Web of Science, Scopus, and ScienceDirect databases (January 2000 – December 2017), based on two search terms: “plantation* + bird*” and “plantation* + avian*”. The literature survey followed the PRISMA statement (Moher et al., 2009). We limited our search to peer-reviewed articles in English, excluding reviews and grey literature (e.g., theses, technical reports, institutional dossiers). An initial search resulted in 1346 articles, which were limited to 291 after filtering irrelevant results (i.e., unrelated articles that matched the search keywords by coincidence). The selected articles were fully read to determine if they met

three criteria: (1) Comparing a native forest stand (i.e., control) with either a native or an exotic plantation of trees (oil palm, coffee, cocoa, banana, and hybrid poplar plantations were not included); (2) Reporting bird species richness or abundance for the native forest and the plantation; (3) Reporting the mean, sample size, and a dispersion measurement of the dependent variables (standard error or standard deviation). Inclusion criteria assessment was performed by one person (GJCV) to avoid a potential inter-reviewer bias. As a result of this screening process, we excluded 254 articles that did not meet our inclusion criteria (being the main exclusion reasons: making comparisons among plantations with no native forest as control, and not presenting standard deviation/error values or the data to calculate those values). Finally, we found 37 usable articles that provided 123 case studies (Fig. 1) from around the world (Fig. S1). Since some papers presented more than one case study, we considered as independent cases those reporting results from different plantations.

2.2. Effect size calculation, model, and moderators

We used Hedges' unbiased standardized mean difference to measure the effect size of each study case (Hedges and Olkin, 1985). This measurement is commonly used to estimate the magnitude of the effect of a particular situation by comparing a control and an experimental group (Gurevitch et al., 2001). As we included studies reporting a variety of species and geographic locations, a random effects model was the appropriate approach to analyze our data (Borenstein et al., 2009). Considering that we gathered more than one case study from some articles, we included article ID as a random factor; therefore, we used mixed-model effects for the analysis of the moderator variables. We conducted separate analyses for species richness and abundance datasets. We estimated the overall effects on both species richness and abundance, and then we defined nine moderator variables: (1) geographical region (tropical or temperate), (2) plantation type (exotic or native), (3) plantation composition (monoculture or mixed plantation), (4) plantation age (young [5–8 years], mature [10–15 years], or old [> 25 years]), (5) plantation size (small [≤ 36 ha] or large [> 50 ha]), (6) plantation purpose (commercial or protective), (7) understory presence (present or absent), (8) landscape surrounding the plantation (connected or isolated), and (9) plantation management (managed or not managed); details about moderator level are available in Tables S1 and S2. Not all case studies reported information for plantation age, size, understory presence, purpose, connectivity or management; therefore, we examined those moderator levels using only the cases with information available (detailed information about moderator level frequencies is available in Figs. S2 and S3). To examine the heterogeneity among moderator levels, we estimated the between-group homogeneity Q_{between} value, which is a χ^2 -distributed statistic that compares the variation between and within moderator levels (Higgins et al., 2003), found to be more appropriate for random-effect models than other heterogeneity measurements, such as I^2 or τ^2 (Borenstein et al., 2009). For either species richness and abundance, we ranked moderator variables in order of importance. Variable ranking was based on three criteria: (a) direction of effect size, (b) magnitude of effect size, and (c) heterogeneity among moderator levels.

2.3. Publication bias

To quantify potential publication bias, we calculated Kendall's tau correlation with continuity correction to determine if effects and sample size were correlated. Since studies reporting null effects are unlikely to be published and hence not included in literature surveys, we estimated Rosenthal's fail-safe number that indicates the number of unpublished case studies with null effects needed to reverse the observed effects (Hillebrand, 2008). This calculation indicates a robust result when the fail-safe number is equal or greater than $5N + 10$, where N is the number of case studies. Despite the criticism on

Rosenthal's fail safe method (see Jennions et al., 2013), we decided to use this classic approach since weighted approaches (e.g., Rosenberg's method) are known to perform poorly on random effect models (Jennions et al., 2013). We also visually examined the relationship between effect and sample size using the funnel plot approach and tested its significance using Egger's test (Egger et al., 1997; Hedges and Vevea, 1996). Then, we conducted a 'trim and fill' procedure (Duval and Tweedie, 2000a, b) that accounts for potential bias related to asymmetries of the positive and negative cases distribution, by recalculating the mean effect and confidence intervals to verify the robustness of the results (Jennions and Møller, 2002). Also, we followed the recommendations of Nakagawa et al. (2017) in order to get informative results by taking into account data independence, publication bias, and the potential effects of outliers. All analyses were conducted using Comprehensive Meta-Analysis 3.0 software (Borenstein et al., 2005).

2.4. Data accessibility

Original data associated to this article is available from the figshare digital repository: <https://doi.org/10.6084/m9.figshare.7473005>

3. Results

3.1. Species richness

Of the 123 case studies obtained from the literature survey, 68 were related to species richness and 55 to abundance (details on Table S3). There was an overall negative effect of forest plantations on species richness (Fig. 2a), which is consistent at tropical and temperate regions (Fig. 2b). The effects of exotic and native plantations were also negative (Fig. 2c), with exotic plantations having more negative effects. However, mixed plantations showed no significant effects, contrary to what is found at monocultures (Fig. 2d). We observed that the negative effects decrease with plantation age (Fig. 2e), but increase with plantation size (Fig. 2f). We also found that commercial plantations had a negative effect on bird species richness, whereas protective plantations had a neutral effect (Fig. 2g). Additionally, we found a negative effect on species richness, regardless of the presence of understory vegetation (Fig. 2h), landscape connectivity (Fig. 2i), or management (Fig. 2j). In order of importance, species richness may be affected by: plantation size, age, purpose, type, origin, landscape connectivity, understory presence, and management.

3.2. Bird abundance

There was a negative effect of forest plantations on the number of individuals (Fig. 3a). This effect was found for tropical and temperate regions (Fig. 3b), but the effects were more pervasive at temperate forests. Exotic tree plantations presented a negative effect on bird abundance, but plantations based on native tree species did not (Fig. 3c). Similarly, mixed plantations showed no effects, whereas monocultures had negative effects (Fig. 3d). We observed that the negative effects decreased with plantation age (Fig. 3e) but increased with plantation size (Fig. 3f). We also found a negative effect of commercial plantations on bird abundance, but a positive effect of plantations established for protective uses (Fig. 3g). Additionally, we found a negative effect of understory vegetation (Fig. 3h), no effects of landscape connectivity (Fig. 3i), and a negative effect of non-managed plantations (Fig. 3j). In order of importance, bird abundance may be affected by: plantation purpose, origin, type, age, size, understory presence, management, and landscape connectivity.

3.3. Publication bias

We found no correlation between effect and sample size for species

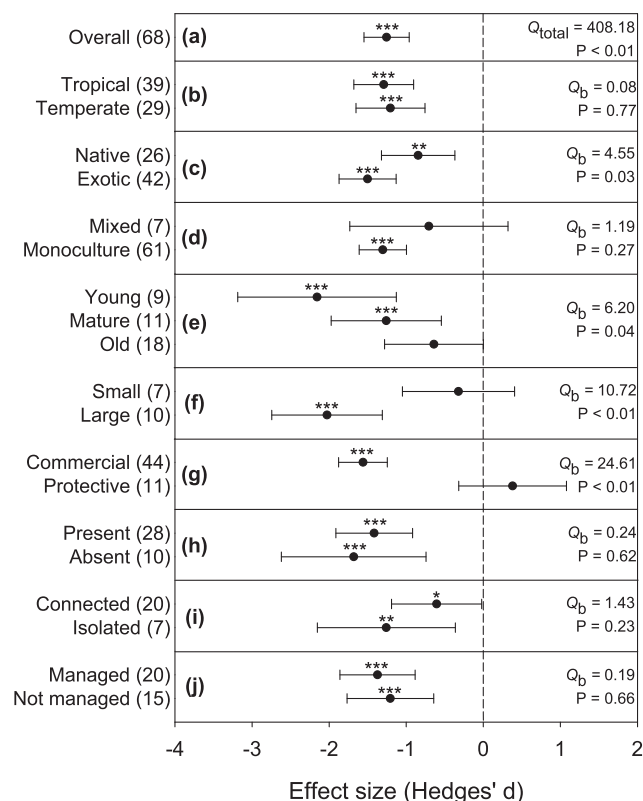


Fig. 2. Effect of forest plantations on bird richness compared to native forests. Mean and 95% confidence intervals are shown for: (a) overall effect, (b) geographical region, (c) plantation origin, (d) plantation type, (e) plantation age, (f) plantation area, (g) plantation purpose, (h) understory presence, (i) landscape, and (j) management. $Q_B = Q_{\text{between}}$ statistics. Significance: $^{NS} P \geq 0.05$, $^* P < 0.05$, $^{**} P < 0.01$, $^{***} P < 0.001$.

richness (Kendall's tau = -0.15 , $P = 0.06$) or abundance (Kendall's tau = -0.12 , $P = 0.20$) data. Rosenthal's fail-safe number criterion indicated that the results are robust and the potential effect of missing unpublished cases is negligible. In this regard, we would need to include 5870 additional cases to change richness results (versus a threshold value of 350 cases), and include 2147 additional cases to change abundance results (versus a threshold of 285 cases). Funnel plots showed an even distribution of positive and negative cases for richness and abundance data (Fig. S4), with no significant asymmetry in any case ($P = 0.06$ for richness and $P = 0.67$ for abundance, after Egger's test). The direction and significance of our results did not change after performing the 'trim and fill' procedure (Table S4), indicating that the results are robust and are not influenced by potential asymmetry bias.

4. Discussion

4.1. General trends

Overall, forestry plantations have negative effects on bird species richness and abundance, with a large variability among case studies. Our results show that certain features of forest plantations may affect bird diversity. Based on many local studies, we made a global assessment aiming to identify general patterns that could make forest plantations friendlier to bird diversity. Some of these features have been previously associated with bird diversity in local studies, but others have not been considered yet. Moreover, other factors, such as the presence of understory and the lack of management, have been pointed as relevant for avifauna diversity in local studies (e.g., Nájera and Simonetti, 2009), but our global analysis showed the opposite.

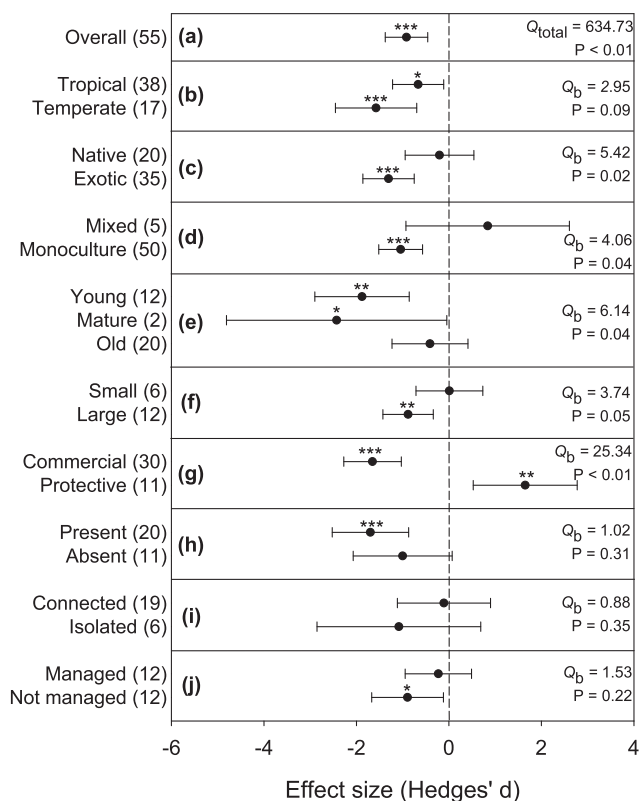


Fig. 3. Effect of forest plantations on bird abundance compared to native forests. Mean and 95% confidence intervals are shown for: (a) overall effect, (b) geographical region, (c) plantation origin, (d) plantation type, (e) plantation age, (f) plantation area, (g) plantation purpose, (h) understory presence, (i) landscape, and (j) management. $Q_B = Q_{\text{between}}$ statistics. Significance: $^{NS} P \geq 0.05$, $^* P < 0.05$, $^{**} P < 0.01$, $^{***} P < 0.001$.

According to our results, the presence of native understory and the absence of management did not improve species richness and abundance, as we initially expected, suggesting that global trends are not necessarily consistent with some local studies. Together, those features could be applied in management plans to improve the biodiversity value of forest plantations, aiming to reduce their negative impacts on native avifauna. Additionally, these results could be used on forest restoration programs to improve fauna recovery on deforested/degraded lands and, therefore, reduce their recovery debt (Meli et al., 2017; Moreno-Mateos et al., 2017). It is important to go beyond local studies and find global patterns related to bird diversity, not only to be able to manage them, but also to make better predictions on patterns of biodiversity loss in a changing world.

4.2. Exotic versus native plantations

In addition to changes in species richness and abundance, we expected changes in species composition between native and planted forests. Exotic forest plantations are usually inhabited by generalist bird species with broad distribution ranges, while endemic, threatened, and forest-dependent species are usually absent from these habitats (e.g., Barlow et al., 2007; Peh et al., 2006). Such composition differences may result from different environmental filters imposed by forest plantations, favoring certain species over others (Castaño-Villa et al., 2019). Native plantations had less negative effects on bird species richness and abundance than plantations based on exotic tree species. Several studies have shown that plantations based on native trees show higher species richness, not only for birds but also for other taxa (e.g., arthropods) that establish trophic relationships with birds (Calviño-Cancela, 2013; Volpato et al., 2010). Unfortunately, most case studies included in this

meta-analysis do not reported species composition data, precluding to make this comparison along with species richness and abundance. In many cases, native plantations show similar bird species richness to that found in native forests (Castaño-Villa et al., 2014a; Proença et al., 2010; Volpato et al., 2010), or even similar species composition (e.g., Farwig et al., 2009). Such similarity may result from the shared evolutionary history between the plantation and the native forest stands, allowing the occurrence of many native species along with their ecological interactions and trophic relationships, which ultimately favors biodiversity. Also, these similarities may result from the landscape context and not from the plantation type *per se* (Fontúrbel et al., 2016). On the other hand, lower bird diversity in exotic plantations could be related to the lack of food resources (particularly insects and fleshy fruits) and nesting cavities (Barlow et al., 2007; Calviño-Cancela, 2013).

Along the same line, mixed forest plantations were less negative than monospecific plantations. Considering that bird diversity is highly sensitive to habitat structure and complexity (Lindenmayer et al., 2006), mixed plantations offer more environmental heterogeneity (Kerr, 1999), as a result of differences in the physiognomy and growth of each planted tree species. From this perspective, a larger structural complexity allows for more diverse bird assemblages (Castaño-Villa et al., 2014b) favoring certain bird groups with particular foraging strategies, such as insectivore birds (Castaño-Villa et al., 2019; Holbech, 2009). We also found those intended for protective use were usually established with native tree species, using mixed arrangements. Contrary to plantations with a commercial purpose, those established to protect watersheds and recover degraded lands (e.g., Castaño-Villa et al., 2014a) are commonly established using native species in mixed arrangements, which indirectly benefits many forest-dependent animals and potentially aid forest recovery (Crouzeilles et al., 2016). Despite the fact that planted forests present lower bird diversity than natural forests, they can be a better alternative than other land uses (e.g., pastures; Felton et al., 2010).

Recently established forest plantations had more negative effects on bird diversity than those established > 25 years ago. Due to recolonizing processes, bird diversity is particularly low in young (< 8 years old) plantations (Calviño-Cancela, 2013; Styring et al., 2011). This pattern could result from the dense and homogeneous canopy structure found in young plantations, which precludes understory vegetation development (Oliver and Larson, 1996). Contrarily, vegetation structure in old plantations is more similar to that of native forest stands (Murcia, 1997), allowing them to hold similar bird communities in a few cases (e.g., Fontúrbel et al., 2016). Therefore, longer rotation times may favor wildlife persistence in forest plantations by allowing the development of a more complex habitat structure (Law et al., 2014; Millan et al., 2015). Also, small (< 36 ha) plantation stands showed to be friendlier than large (> 50 ha) plantations, probably due to edge and core area effects (Didham and Ewers, 2012; Laurance and Yensen, 1991). Further, these plantation features are non-independent from each other. We can distinguish two major feature associations in our study: (1) plantations based on exotic species are mostly established as monocultures, intended for commercial purposes, actively managed, and established in large (> 50 ha) stands; and (2) plantations based on native species are either established as monocultures or mixed plantations, intended for protective uses, not actively managed, and established in small (< 36 ha) stands.

4.3. Landscape context

At a broader level, heterogeneous landscape mosaics favor the colonization of plantation stands by many species (e.g., plants and insects) that come from the neighboring habitats (i.e., native forest remnants). Such heterogeneous landscape mosaics may provide complementary resources for birds (e.g., food, shelter) that allow them to use forest plantations as habitat (Laurance and Bierregaard, 1997). Hence, those

bird species that are able to use these resources may be favored by landscape heterogeneity (Estades et al., 2012), but the benefit for forest-specialist birds might be negligible. Whereas landscape heterogeneity may improve species richness, landscape connectivity is expected to affect bird abundance as it usually decreases when plantations are isolated from native forest remnants (Farwig et al., 2009).

Landscape ecology theory states that isolation among habitat remnants precludes the colonization from source habitats; therefore we expected a larger bird species richness and abundance when forest plantations and native remnants are connected (Kavanagh et al., 2007). However, the colonization probability is affected not only by the distance among habitat remnants, but also by the ability of forest birds to disperse through non-forested habitats (Castillo-Escrivà et al., 2018; Sieving et al., 1996). Despite landscape connectivity has been identified as a key factor to enhance biodiversity in forest plantations (Kavanagh et al., 2007; Lindenmayer and Hobbs, 2004; Lindenmayer et al., 2010), our results show otherwise. While the effects on species richness are less negative on connected plantations compared to isolated ones, we observed no effects on bird abundance in any case. These results suggest that landscape connectivity is critical only for dispersal-restricted understory birds (e.g., rhynocryptids; Sieving et al., 1996), but most bird species would not be affected by spatial discontinuities as their flying capabilities allow them to disperse long distances through non-forested matrices.

4.4. Concluding remarks

Our global assessment showed that, although the effects of forest plantations on bird diversity are highly variable, some general patterns emerge. Given that forest plantations are structurally simpler than native forests and are dominated by one or a few canopy species, they negatively affect both bird species richness and abundance. However, not all plantations are the same: those with the least negative effects are based on native tree species, in mixed arrangements, aimed for protective purposes, established in small stands with long rotation times, and connected to native forest remnants. On the other hand, plantations based on monocultures of exotic species for commercial purposes, established in large stands with short rotation times, showed to be the most negative for birds. In any case, forest plantations cannot replace native forests and, as even with similar species richness and abundance values, bird community composition will be dissimilar as result of the environmental filters imposed by each habitat type (Castaño-Villa et al., 2019). Therefore, forest plantations could be managed using features to enhance bird diversity by (i) allowing bird colonization, (ii) generating a more complex vegetation structure that provides food and shelter resources, benefiting not only generalist bird species but also those more specialized to forest habitats, and (iii) promoting complex productive landscapes. These management guidelines could provide an additional value to forest plantations, besides timber production, by mitigating the negative effects on the associated biodiversity, with a long-term perspective.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2019.03.025> and <https://doi.org/10.6084/m9.figshare.7473005>.

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